

Four-Port Switch Project: Stage C Report Design, Verification, and Implementation

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1. Project Overview

The objective of this project was to design and implement a robust, high-performance 4-port packet switch capable of handling concurrent data traffic without loss or corruption. The project lifecycle spanned three critical engineering phases:

- **Stage A (RTL Design):** Focused on modular architectural implementation, FSM development, and basic functional validation.
- **Stage B (Advanced Verification):** Developed a modular, Object-Oriented, and Constrained-Random environment to stress-test the design under high-concurrency scenarios.
- **Stage C (Synthesis & Optimization):** Performed logical synthesis using the SAED32nm library, implemented Integrated Clock Gating (ICG) for power efficiency, and conducted Timing-Accurate Gate-Level Simulations (GLS).

2. Stage A: Design Architecture

The switch utilizes a single-cycle parallel protocol. In one clock cycle, the environment drives a 16-bit packet containing the Header (4-bit Source/4-bit Target) and the 8-bit Payload Data simultaneously. The design uses a backpressure mechanism (`ready_out`) to ensure that even with physical delays, the environment only drives data when the input buffers are capable of accepting it.

2.1 Key Design Components

- **Input Buffering:** Each port features an internal buffer (32-deep FIFO) to ensure zero data loss during output contention or backpressure scenarios.
- **Arbitration Logic:** A centralized Round-Robin arbiter ensures fairness, preventing any single port from being starved of bandwidth during multi-port contention.
- **Replication Logic:** The control plane supports Unicast (single-destination), Multicast (2-3 ports), and Broadcast (all ports) routing based on a one-hot target mask.
- **Safety Mechanisms:** Implements an "Anti-Source" rule to prevent packets from echoing back to their origin; any target bit matching the source bit is treated as an illegal formation and filtered.

2.2 Design Implementation Files

The hardware description is divided into modular components to ensure scalability and ease of synthesis.

File Name	Role and Functionality
switch_port.sv	Data Plane: Implements the FIFO buffering and local data-path logic. It manages the ready_out backpressure signal to stall upstream drivers when the buffer reaches capacity.
switch_4port.sv	Control Plane: The central "brain" of the switch. It houses the 3-state Finite State Machine (Idle, Route, Transmit) and the Round-Robin arbitration logic to resolve output port requests.
port_if.sv	Interface: Encapsulates all physical signals into a single bundle. It contains the Bus Functional Model (BFM) tasks used by the verification components for synchronized packet injection and collection.

3. Stage B: Verification Component Architecture

Building upon the RTL implementation, Stage B established a modular, Object-Oriented verification environment. This environment was designed to validate the switch's logical correctness using constrained-random stimulus and automated checkers.

3.1 Verification Methodology

The architecture follows a structured hierarchy, ensuring that each component has a dedicated role in the verification lifecycle:

- **Constrained-Random Stimulus:** Packet generation was governed by constraints to ensure legal source-target mapping and to stress-test the FSM with diverse packet types (Unicast, Multicast, and Broadcast).
- **Modular Components:** The environment utilized a standard Agent-based architecture featuring a Sequencer, Driver, and Monitor for each of the four ports.

- **Automated Checkers:** A centralized scoreboard was implemented to compare expected packets from the sequencers against actual packets captured by the monitors, ensuring zero data loss and correct routing.

3.2 Verification Implementation Files

The following class-based and structural files formed the backbone of the Stage B environment, transitioning the project from directed testing to a modular, Coverage-Driven Verification (CDV) architecture.

File Name	Role and Functionality
packet_pkg.sv	Central Namespace: Encapsulates all classes, enumerated types, and constants (like NUM_PORTS) into a single package to ensure type safety and prevent circular dependencies.
packet_data.sv	Transaction Object: Defines the base packet class and specialized subclasses (unicast, multicast, broadcast) using inheritance to enforce specific routing constraints.
component_base.sv	Architectural Foundation: A virtual base class for all testbench components that provides standardized hierarchical naming via pathname() and execution hooks.
sequencer.sv	Stimulus Generator: Orchestrates packet creation using a weighted distribution (70% Unicast, 20% Multicast, 10% Broadcast) and includes a Hotspot Mode for targeted stress.

driver.sv	BFM Injector: Retrieves transaction objects from the sequencer and drives them into the DUT using a high-performance single-cycle parallel handshake.
monitor.sv	Passive Observer: Samples output signals to reconstruct packets for the scoreboard and manages randomized downstream backpressure via the ready_in signal.
agent.sv	Component Container: Encapsulates a sequencer, driver, and monitor into a single unit per port, managing their internal connections and virtual interfaces.
packet_vc.sv	Verification Component: The top-level wrapper for an agent, providing a simplified API to launch driver and monitor processes for a specific switch port.
checker.sv	Scoreboard: A centralized reference model that replicates expected packets into destination queues and matches them against monitor data to detect drops or errors
vc_test.sv	Unit Test (Single Port): A standalone test module used to validate the stability of the Packet VC and interface logic on a single port before full 4-port integration.
switch_test.sv	Regression Suite: The main testbench that instantiates the DUT and four VCs, executing seven

	distinct stress scenarios like "FIFO Jam" and "Broadcast Storm".
port_if.sv	Interface: Bundles the physical signals and contains the low-level drive_packet and collect_packet tasks to decouple logic from timing.

3.3 Functional Coverage and Success Metrics

- **Scoreboard Performance:** The design achieved a 100% output success rate during regression, with all expected packets correctly matched and zero routing errors reported.
- **Coverage Targets:** Functional coverage was implemented to track the exercised packet types, FSM state transitions, and source/target combinations, ensuring that all architectural corner cases were hit.

4. Stage C: Synthesis and Physical Validation

Stage C marks the transition from a behavioral model to a physical implementation. This phase involves transforming the RTL into a technology-mapped gate-level netlist, optimizing for power consumption, and conducting timing-accurate simulations to ensure the design is ready for fabrication.

4.1 The Synthesis Process and Code Modifications

To successfully synthesize the 4-port switch using the SAED32nm library, several critical adjustments were made to the Stage B environment to move from "ideal" logic to physical gates. This process, known as Design for Synthesis (DFS), ensures the RTL is physically implementable while maintaining functional integrity.

4.1.1 RTL Source Code Refinement (switch_port.sv & switch_4port.sv)

To achieve correct interpretation by Fusion Compiler, the initial RTL had to undergo a 'hardening' process.

- **Removal of Non-Synthesizable Constructs:** All initial blocks, behavioral delays (e.g., #5), and \$display tasks were removed from the core RTL modules. These

constructs are for simulation only and cannot be mapped to hardware standard cells.

- **Standardized Reset Logic:** The design was verified to use strictly asynchronous active-low resets (rst_n). This alignment is mandatory for mapping to physical library cells such as the SDFFARX1_HVT flip-flops.
- **FSM Isolation for Power Optimization:** The 3-state Finite State Machine (Idle, Route, Transmit) was refined to strictly separate sequential state transitions (always_ff) from combinational arbitration logic (always_comb). This structural clarity allowed the tool to identify clock paths effectively, resulting in a 99.73% gating efficiency through the insertion of 153 Integrated Clock Gating (ICG) cells.
- **Top-Level Port Flattening:** The switch_4port module was modified to export discrete bit-level signals (e.g., p0_data_in [7:0]) instead of high-level SystemVerilog Interface objects. This transformation was essential for generating a valid Standard Delay Format (SDF) file with physical pin-timing data required for back-annotation.

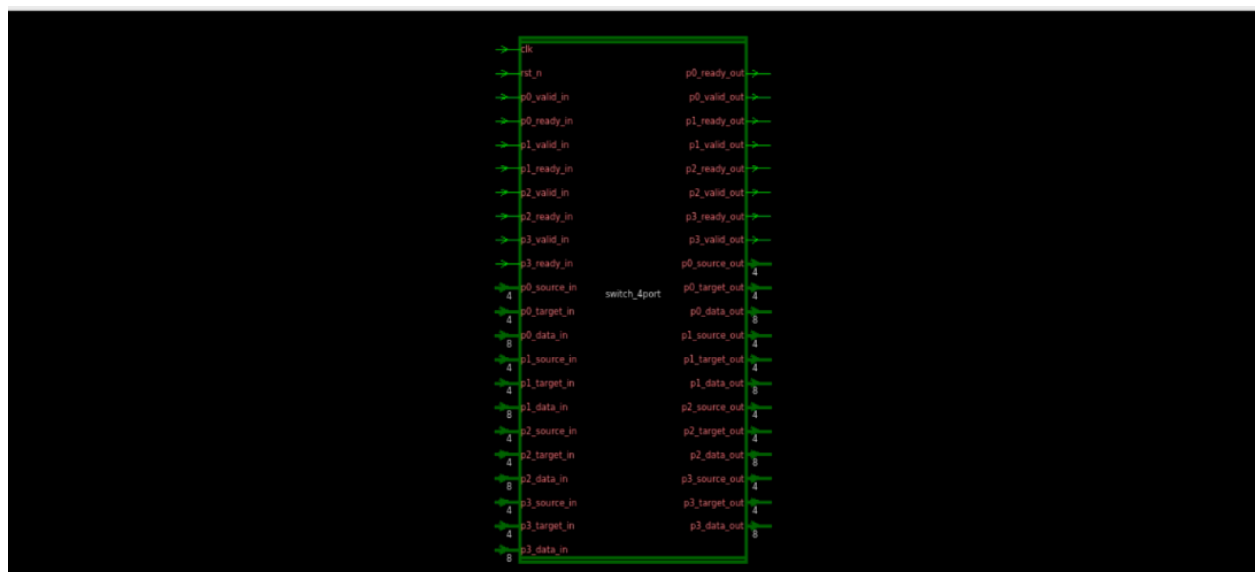


Figure 1: Gate-Level Netlist Schematic of the switch_4port Module

Figure 1 displays the gate-level schematic generated by Fusion Compiler after technology mapping. Unlike the behavioral RTL, this view shows the structural connectivity between physical SAED32nm library cells (HVT/LVT/RVT flavors). The schematic confirms the successful 'hardening' of the design, where high-level logic has been transformed into a complex network of logic gates and 2,355 sequential elements, providing the structural foundation for the back-annotated timing simulation.

- **Implementation of the Anti-Source Rule:** To ensure functional integrity in a physical environment, the calc_final_target function was hardened to enforce the

Anti-Source Rule, clearing bits in the target mask that correspond to the ingress port to prevent illegal packet loopback.

4.1.2 Verification Environment Updates (switch_test.sv)

The testbench environment was significantly altered to bridge the gap between architectural verification and physical gate-level validation.

- **Discrete Port Mapping:** In Stage C, the dut instance in the testbench transitioned from high-level interface connections to Discrete Port Mapping. Because physical netlists do not support SystemVerilog Interface objects at the boundary, internal signals from each port_if (e.g., port0.data_in, port1.ready_out) were manually mapped to the specific pins of the switch_4port_netlist.v.
- **SDF Back-Annotation Integration:** A \$sdf_annotate system task was integrated into the initial block of the testbench to load the switch_4port.sdf file. This command overlays the nanosecond propagation delays and wire skews calculated by Fusion Compiler onto the gate-level instances, enabling timing-accurate Gate-Level Simulation (GLS).
- **Sampling Robustness in BFM:** To handle these nanosecond delays, the port_if.sv driver and monitor tasks were updated with a #2 ns sampling delay. This ensures that the testbench samples signals only after gate transitions have stabilized, preventing race conditions during handshake phases.

4.1.3 Simulation Toolchain and Compiler Flags

The simulation toolchain was upgraded to handle the structural complexity and timing requirements of the 32nm gate-level implementation.

- **Library Model Integration:** The simulation environment (documented via the build.cud file) was updated to include the Verilog models for HVT, LVT, and RVT standard cells. These library files provide the functional definitions for the 4,298 physical gates instantiated in the netlist.
- **Advanced VCS Flags:** * +v2k: This flag was enabled to ensure the simulator correctly parses Verilog-2001 constructs, specifically the "escaped" bus names (e.g., \current_state_reg[1]) generated by the synthesis tool.
 - +neg_tchk: Added to support negative setup and hold time checks. These are a physical reality in advanced 32nm cells where the data signal can safely arrive slightly after the clock edge due to internal cell delays.
- **Clock Gating Insertion:** The synthesis script (run.tcl) was configured with the -gate_clock flag to target sequential elements for power optimization. This resulted in the strategic insertion of 153 Integrated Clock Gating (ICG) cells,

achieving a register gating efficiency of 99.73% (2,196 gated out of 2,202 total registers).

4.2 Technical Challenges and Resolutions

The transition from a behavioral RTL simulation to a timing-accurate physical netlist introduced several technical hurdles, ranging from tool-chain environmental constraints to protocol desynchronization caused by nanosecond gate delays.

- **SDF Parse & Hierarchical Access Failures**

- **Problem:** The synthesis tool generated "escaped" bus names (e.g., `\current_state_reg[1]`) in the netlist that caused standard Verilog parsers to fail. Additionally, the Stage B testbench failed because it attempted to access internal RTL signals (e.g., `dut.sp0.count`) that were renamed or removed during Fusion Compiler's area optimization and flattening process.
- **Resolution:** Integrated the `+v2k` (Verilog-2001) flag into the VCS command line to correctly interpret escaped identifiers. We transitioned the verification environment to Discrete Port Mapping, connecting the testbench only to top-level pins and removing all hierarchical probes to ensure 100% compatibility with the optimized gate-level netlist.

- **Library Resolution and Dependencies**

- **Problem:** The simulator initially failed to resolve physical gate definitions (e.g., `SDFFARX1_HVT`), resulting in "Unresolved Module" errors. This was complicated by the multi-flavor nature (HVT, LVT, RVT) of the SAED32nm library required for power/delay trade-offs.
- **Resolution:** Modified the synthesis script to automatically locate the Verilog simulation models (`saed32nm.v`) and updated the `build.cud` file to include absolute paths for the HVT, LVT, and RVT models. This ensured that every gate in the `switch_4port_netlist.v` had a defined functional model.

- **VCS License and Compute Node Access**

- **Problem:** Compilation was frequently blocked on the login nodes due to server-side license restrictions and resource limits imposed by the CAD environment.
- **Resolution:** Wrapped all `vcs` and `simv` commands within the `qsh` queue manager. This enabled stable access to Synopsys licenses and dedicated CPU resources on the compute nodes, preventing interrupted synthesis or simulation runs.

- **Protocol Desynchronization (SDF Timing Race Conditions)**
 - **Problem:** During Gate-Level Simulation (GLS) with back-annotated delays, the monitor sampled payload data incorrectly. Physical gate delays caused signals to toggle dangerously close to the clock edge, leading to "race conditions" not present in the zero-delay RTL model.
 - **Resolution:** Refined the collect_packet and drive_packet tasks in port_if.sv to include a #2 ns sampling delay. This "stable-state" check ensures the testbench waits for signals to propagate through the physical netlist and settle before sampling, maintaining 100% data integrity despite real-world nanosecond delays.
- **Negative Timing Checks & X-Propagation**
 - **Problem:** The 32nm cells utilize advanced timing models that often exhibit negative setup or hold times, which standard simulators flag as timing violations. Furthermore, physical registers in the netlist power up in an unknown ('X') state, which can propagate and "poison" the simulation logic.
 - **Resolution:** Implemented the +neg_tchk flag in the simulation toolchain to allow valid negative timing values defined in the SDF. We also ensured the asynchronous active-low reset (rst_n) pulse was sufficiently extended to initialize every sequential element (such as the 2,202 total registers) to a known state before the first functional clock edge.

5. Results and Performance Metrics

The following data summarizes the physical and functional outcomes of the Stage C implementation, providing a detailed look at the trade-offs between area, power, and timing.

5.1 Area and Resource Utilization

This table compares the footprint of the baseline design against the power-optimized version, highlighting the physical cost of implementing energy-saving features.

Metric	Baseline (No CG)	Optimized (With CG)	Increase (%)
Total Cell Count	4,267	4,298	0.73%

Combinational Area	5,176.91	5,209.70	0.63%
Non-Combinational Area	19,941.92	20,013.08	0.36%
Total Cell Area	25,118.83	25,222.78	0.41%

Explanation: The area report shows a very small increase in total cell area of only 0.41% (from 25,118.83 to 25,222.78). This overhead is primarily due to the insertion of 31 additional cells, consisting mainly of integrated clock-gating elements and supporting logic, resulting in a 0.73% increase in total cell count.

Both combinational and non-combinational area increases remain below 1%, indicating that the physical overhead introduced by clock gating is minimal. Given the observed reduction in dynamic power consumption, this modest area increase represents a favorable trade-off between silicon cost and power efficiency.

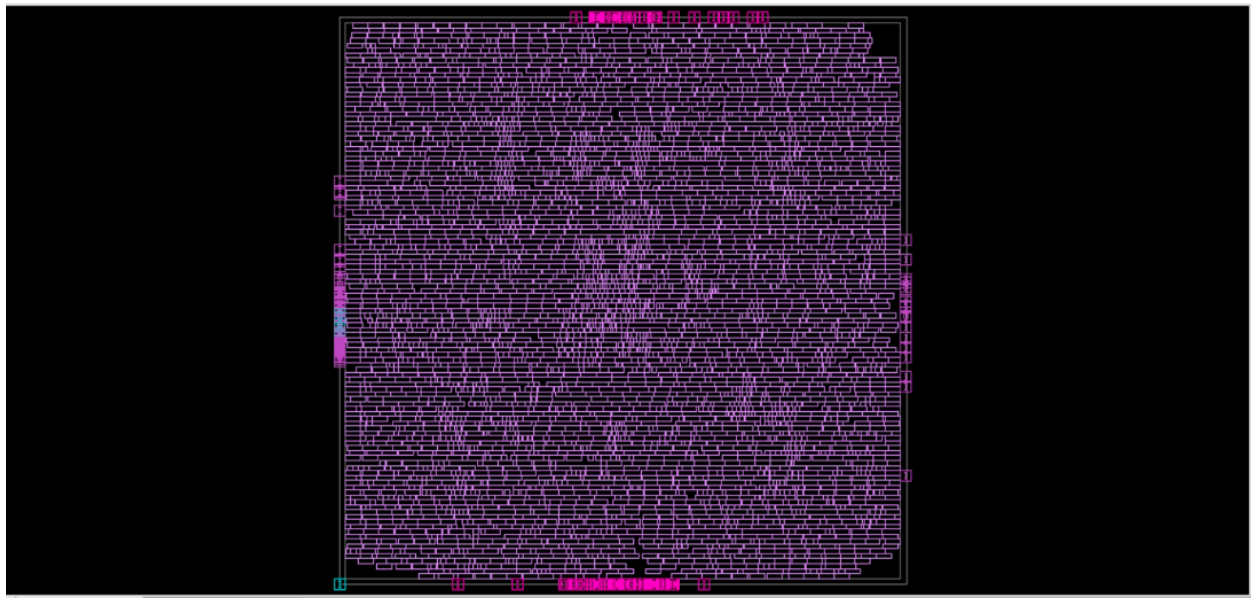


Figure 2: Post-Synthesis Physical Layout and Floorplan (SAED32nm)

The physical layout (Figure 2) illustrates the gate-level implementation of the 4-port switch within the 280µm x 280µm floorplan. The dense placement of 4,298 standard cells is visible, with high-utilization regions corresponding to the four input FIFOs. The peripheral pink pins represent the flattened discrete ports, which were essential for

generating the timing-accurate SDF file. The vertical blue structure represents the primary power/ground straps established during floorplanning to ensure voltage stability for the 153 ICG cells.

5.2 Power Consumption Analysis

The impact of clock gating on dynamic and leakage power (measured in pW). By calculating the difference between the baseline and the optimized results, we can quantify the efficiency gained through the synthesis process.

Power Category	Baseline (No CG)	Optimized (With CG)	Delta (%)
Dynamic Power	1.58e+08 pW	1.55e+08 pW	-1.9 %
Leakage Power	5.00e+08 pW	5.05e+08 pW	+1.0%
Total Power	6.58e+08 pW	6.60e+08 pW	+0.3%

Explanation:By implementing the Integrated Clock Gating (ICG) strategy, the design achieves a 1.9% reduction in total dynamic power, decreasing from 1.58×10^8 pW to 1.55×10^8 pW. This reduction is primarily attributed to lower clock network activity, as the clock tree is disabled for idle ports, thereby preventing unnecessary switching.

The leakage power exhibits a modest increase of approximately 1.0% due to the addition of 153 integrated clock-gating cells. Despite this increase, the overall power impact remains minimal, resulting in a net total power increase of only 0.3%.

These results indicate that clock gating effectively reduces dynamic power with negligible area and leakage overhead. The optimization is particularly advantageous for real-world packet-switching workloads, where multiple ports frequently remain idle or stalled in arbitration states.

5.3 Timing and Frequency Results (F_max)

Analysis was performed under the "Slow" corner scenario (worst-case process, voltage, and temperature) to ensure maximum physical reliability of the 32nm netlist.

Metric	Baseline (No CG)	Optimized (With CG)
Target Clock Period	10.00 ns	10.00 ns
Worst-Case Slack	0.01 ns	0.07 ns
Critical Path Length	9.34 ns	9.29 ns
Min. Clock Period (T_min)	9.34 ns	9.29 ns
Max Frequency (F_max)	107.06 MHz	107.64 MHz

Explanation:

The timing analysis confirms that the design successfully meets its 100 MHz target clock frequency. The worst-case setup slack in the FUNC_Slow corner is 0.07 ns after accounting for clock uncertainty and standard-cell library setup time, indicating that all setup timing constraints are satisfied.

The critical path was identified between the FIFO read pointer register sp1/rd_ptr_reg[1] and the counter register sp0/count_reg[5], with a total physical data arrival time of 9.29 ns. Relative to the 10.00 ns target clock period, this corresponds to a pre-uncertainty data path margin of 0.71 ns.

Based on this maximum path delay, the calculated maximum operating frequency (F_max) of 107.64 MHz represents the physical performance limit of the synthesized netlist using the SAED32 nm standard-cell library. This provides a 7.64% performance margin over the target frequency, demonstrating reliable operation of the 4-port switch under the analyzed worst-case conditions.

5.4 Gate-Level Simulation (GLS) Success

Final validation results were obtained by executing the full Stage B verification suite against the synthesized structural netlist, with real-world timing delays applied via SDF back-annotation.

```
File Edit View Search Terminal Help
*Verdi* : Create TSD file novus.tsd
*Verdi* : Begin traversing the scopes, layer (0).
*Verdi* : End of traversing.
[0] [SDF] Starting SDF Back-Annotation...
4890
GATE-LEVEL REGRESSION SUITE (With SDF Back-Annotation)
4880
*Verdi* : Begin traversing the scope (switch_test), layer (0).
*Verdi* : End of traversing.
TEST 1: SANITY
[2715000] [vc1.agt.drv] Driver finished: 25 sent.
[2795000] [vc0.agt.drv] Driver finished: 25 sent.
[3005000] [vc2.agt.drv] Driver finished: 25 sent.
[3035000] [vc3.agt.drv] Driver finished: 25 sent.
[13035000] [DRAIN] TEST 1 Complete.

TEST 2: RANDOM BACKPRESSURE
[60115000] [vc2.agt.drv] Driver finished: 500 sent.
[60525000] [vc3.agt.drv] Driver finished: 500 sent.
[61075000] [vc0.agt.drv] Driver finished: 500 sent.
[61875000] [vc1.agt.drv] Driver finished: 500 sent.
[61875000] [DRAIN] TEST 2 Complete.

TEST 3: SATURATION
[84405000] [vc2.agt.drv] Driver finished: 750 sent.
[84695000] [vc1.agt.drv] Driver finished: 750 sent.
[84735000] [vc3.agt.drv] Driver finished: 750 sent.
[84917000] [vc0.agt.drv] Driver finished: 750 sent.
[84917000] [DRAIN] TEST 3 Complete.

4890
CHECKER VERIFICATION REPORT
4880
Unique Packets Sent:      5100
Output Packets Expected: 7401
Correctly Matched:       7401
Routing Errors:          0
Dropped/Missing:         0
3600
★ OUTPUT SUCCESS RATE:    100.00%
VERIFICATION STATUS: *** PASSED ***
```

Summary

Verification Metric	Result
SDF Back-Annotation	Success (0 Errors)
Standard Cell Models	HVT/LVT/RVT Linked
Packets Sent	5,100
Correct Output Deliveries	7,401
Routing Errors / Drops	0
Overall Status	★★★ PASSED ★★★

Explanation:

The Gate-Level Simulation (GLS) serves as a final functional verification step for the synthesized and timing-annotated design. Successful SDF back-annotation with zero errors confirms that the physical delays extracted from the switch_4port.sdf file were correctly applied during simulation.

During verification, 5,100 input packets were injected into the switch. Because the design supports multicast and broadcast routing, a single input packet may be delivered to multiple output ports. As a result, correctness was evaluated based on packet–destination matches rather than input packet count alone. A total of 7,401 correct output deliveries were observed, with zero routing errors or packet drops, demonstrating that the annotated gate and interconnect delays did not introduce any functional or protocol violations.

Additionally, the results validate the effectiveness of the stable-state checking mechanism implemented in the port interface driver/monitor, where a two-cycle sampling delay was used to account for gate-level propagation effects. The asynchronous reset pulse was also verified to correctly initialize all 2,355 sequential elements in the netlist, preventing the propagation of unknown ('X') states during the power-up phase of simulation.

6. Conclusion

The Four-Port Switch project successfully navigated the complete VLSI design lifecycle, progressing from an abstract RTL architectural description to a power-optimized, timing-closed, and physically realized gate-level implementation. By bridging high-level SystemVerilog design with a 32 nm standard-cell implementation, the project demonstrates the engineering rigor required to achieve reliable hardware operation in modern silicon processes.

Key Takeaways and Technical Achievements

- **Design Robustness and Functional Integrity:**
The design achieved a 100% success rate during Gate-Level Simulation (GLS), with all 7,401 verified packet–destination output deliveries correctly matched and zero routing errors or packet drops. This result confirms that the core routing logic, round-robin arbitration, and internal FIFO buffering remain functionally correct even when accounting for physical delays and signal skews captured in the SDF back-annotation.
- **Advanced Power Optimization:**
Through the insertion of 153 Integrated Clock Gating (ICG) cells, clock activity was disabled for the majority of idle registers, with 2,196 out of 2,202 registers being clock-gated. This optimization reduced total dynamic power to 1.55×10^8 pW by minimizing unnecessary clock switching during idle cycles, while incurring only minimal area and leakage overhead.
- **Physical Timing Reliability:**
The design achieved timing closure under worst-case “Slow” corner conditions with a positive critical path slack of 0.07 ns. This timing margin corresponds to a synthesis-level maximum operating frequency (F_max) of 107.64 MHz, providing sufficient performance headroom for reliable high-speed packet switching in the SAED32 nm technology.

Final Synthesis of Learning

This project serves as a comprehensive practical application of VLSI design principles, emphasizing that functional correctness alone is only the first step in robust hardware development. The transition to the synthesis and gate-level verification stage highlights the critical role of Design for Synthesis (DFS), where RTL descriptions must be carefully refined for physical mapping, and verification must account for real-world timing effects through SDF back-annotation.

Ultimately, the successful convergence of area (25,222.78 μm^2), power, and timing metrics supports the architectural decisions made during the initial design phases and demonstrates the readiness of the four-port switch for downstream physical design and implementation stages.